

Further Statistics 1

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Chapter 6: Chi-Squared Tests

Why this chapter matters

Every hypothesis test you have met so far has tested a single numerical parameter (a proportion, a mean, a Poisson rate). The chi-squared (χ^2) test is different in kind: it tests whether an entire set of observed frequencies is consistent with a proposed model, or whether two categorical variables are associated with each other. It is the workhorse test for categorical data and appears in two closely related forms: goodness-of-fit tests and tests for association (contingency tables).

Concept: the chi-squared goodness-of-fit test

A goodness-of-fit test compares a set of observed frequencies (O) to the frequencies expected (E) under a proposed distribution (H_0). The test statistic is:

$$\chi^2 = \sum [(O - E)^2 / E]$$

Under H_0 , this statistic approximately follows a χ^2 distribution with degrees of freedom $v = (\text{number of categories}) - 1 - (\text{number of parameters estimated from the data})$.

Why $(O - E)^2/E$, and not just $O - E$? Squaring makes every discrepancy positive, so an under-count in one category can't cancel an over-count in another when you sum across categories — exactly the same reasoning that motivated squaring in the variance formula back in Chapter 1. Dividing by E then *scales* each squared discrepancy relative to the size of frequency expected there: being out by 5 matters enormously if only 6 were expected, but is barely noticeable if 500 were expected. Each individual term $(O - E)^2/E$ behaves like a squared, standardised deviation, and χ^2 is built by summing several such terms — which is precisely why the total follows a χ^2 (chi-squared) distribution: a χ^2 distribution with v degrees of freedom is, by definition, the distribution of a sum of v independent squared standard Normal variables.

Key vocabulary

Term	Meaning
Observed frequency, O	The actual count recorded in each category
Expected frequency, E	The count predicted by H_0 for each category
Degrees of freedom, v	Controls the shape of the χ^2 distribution used; found by counting categories and subtracting constraints
Pooling / combining classes	Merging adjacent categories with small expected frequencies (each E should be ≥ 5)
Critical value	The value of χ^2 from tables beyond which H_0 is rejected

Key formula: chi-squared test statistic and degrees of freedom
KEY FORMULA

Test type	Degrees of freedom ν
Goodness-of-fit, no parameters estimated	(number of categories) – 1
Goodness-of-fit, k parameters estimated from data	(number of categories) – 1 – k
Contingency table, r rows and c columns	$(r - 1)(c - 1)$

A key rule before calculating X^2 : every expected frequency should be at least 5. If any $E < 5$, adjacent categories must be pooled (combined) until this condition is satisfied, and ν is recalculated using the reduced number of categories.

Procedure: carrying out a goodness-of-fit test

1. State H_0 (data follows the proposed distribution) and H_1 (data does not follow it).
2. Calculate the expected frequency for each category under H_0 (estimating any unknown parameters from the data if required).
3. Check every $E \geq 5$; pool categories if necessary and adjust the number of categories accordingly.
4. Calculate $X^2 = \sum[(O-E)^2/E]$.
5. Find ν and the critical value of χ^2 from tables at the given significance level.
6. Compare X^2 to the critical value: if X^2 exceeds it, reject H_0 . Conclude in context.

Worked example 1: goodness-of-fit test for a given distribution
EXAMPLE 6.1

Question. A die is rolled 120 times with the following results:

Score	1	2	3	4	5	6
Frequency	15	24	17	22	27	15

Test, at the 5% significance level, whether the die is fair.

Step 1. H_0 : the die is fair (each score equally likely, $p = 1/6$ each); H_1 : the die is not fair. **(B1)**

Step 2. Expected frequency for each score = $120 \times 1/6 = 20$. All $E = 20 \geq 5$, so no pooling is needed. **(M1)**

Step 3. Calculate $X^2 = \sum(O-E)^2/E = (15-20)^2/20 + (24-20)^2/20 + (17-20)^2/20 + (22-20)^2/20 + (27-20)^2/20 + (15-20)^2/20 = 25/20 + 16/20 + 9/20 + 4/20 + 49/20 + 25/20 = 128/20 = 6.4$. **(M1A1)**

Step 4. Degrees of freedom $\nu = 6 - 1 = 5$ (no parameters estimated). Critical value $\chi^2_{5,0.05} =$

11.070. (B1)

Step 5. Since $6.4 < 11.070$, X^2 does not exceed the critical value. (M1)

Answer: There is insufficient evidence, at the 5% level, to reject H_0 ; the data is consistent with the die being fair. (A1)

EXAM TIP

 Write out every individual $(O-E)^2/E$ term before summing — not only does this make arithmetic errors easy to spot and correct, but many mark schemes award a method mark for the correctly set-up sum even if the final total is wrong. Also read χ^2 tables carefully: unlike the Normal table, you read along a “degrees of freedom” row and down a significance-level column, and mixing these up is a very common slip.

Concept: goodness-of-fit test with an estimated parameter

 When a parameter of the proposed distribution (e.g. λ for a Poisson model) is not given but must be estimated from the sample data itself, one extra degree of freedom is lost for each parameter estimated.

Worked example 2: goodness-of-fit test for a Poisson model with estimated λ
EXAMPLE 6.2
Question. The number of accidents per week at a factory was recorded over 100 weeks:

Accidents	0	1	2	3	≥ 4
Frequency	38	36	19	5	2

 Test, at the 5% level, whether a Poisson distribution is a suitable model, using the sample mean as an estimate of λ .

Step 1. H_0 : a Poisson distribution is a suitable model; H_1 : it is not. Estimate λ from the data: $\hat{\lambda} = [0(38) + 1(36) + 2(19) + 3(5) + 4(2)]/100 = [0 + 36 + 38 + 15 + 8]/100 = 97/100 = 0.97$. (M1A1)

Step 2. Find Poisson probabilities with $\lambda = 0.97$ and hence expected frequencies ($\times 100$): $P(0) \approx 0.3791 \rightarrow E_0 \approx 37.91$; $P(1) \approx 0.3677 \rightarrow E_1 \approx 36.77$; $P(2) \approx 0.1783 \rightarrow E_2 \approx 17.83$; $P(3) \approx 0.0577 \rightarrow E_3 \approx 5.77$; $P(\geq 4) = 1 - \text{sum of above} \approx 0.0172 \rightarrow E_4 \approx 1.72$. (M1A1)

Step 3. Check $E \geq 5$: the last category ($E \approx 1.72$) is too small, so pool “3” with “ ≥ 4 ” to give a combined category “ ≥ 3 ” with $O = 5 + 2 = 7$ and $E \approx 5.77 + 1.72 = 7.49$. (M1)

Step 4. Calculate $X^2 = (38 - 37.91)^2/37.91 + (36 - 36.77)^2/36.77 + (19 - 17.83)^2/17.83 + (7 - 7.49)^2/7.49 \approx 0.0002 + 0.0161 + 0.0768 + 0.0321 = 0.1252$. (M1A1)

Step 5. Number of categories after pooling = 4; one parameter (λ) estimated, so $v = 4 - 1 - 1 = 2$. Critical value $\chi^2_{2,0.05} = 5.991$. (B1)

Step 6. Since $0.1252 < 5.991$, do not reject H_0 . (M1)

Answer: There is insufficient evidence, at the 5% level, to reject the Poisson model; a Poisson

distribution appears to be a suitable model for the number of accidents per week. **(A1)**

EXAM TIP

This is the question type that catches out otherwise strong students, because THREE separate adjustments compound: estimating λ from data, pooling small expected frequencies, and losing an extra degree of freedom for the estimated parameter. Treat the three steps as a strict checklist rather than something to remember in one go, and always calculate degrees of freedom LAST, after pooling is finalised — pooling changes the number of categories that v is based on.

Concept: chi-squared tests for association (contingency tables)

A contingency table cross-classifies data by two categorical variables. The chi-squared test for association tests H_0 : the two variables are independent, against H_1 : the two variables are associated. Expected frequencies are calculated using:

$$E = (\text{row total} \times \text{column total}) / \text{grand total}$$

for each cell, and degrees of freedom are $v = (\text{rows} - 1)(\text{columns} - 1)$.

Worked example 3: chi-squared test for association

EXAMPLE 6.3

Question. A survey asked 200 people their preferred mode of commuting and whether they live in an urban or rural area:

	Car	Public transport	Cycling/walking	Row total
Urban	40	50	30	120
Rural	50	20	10	80
Column total	90	70	40	200

Test, at the 1% significance level, whether commuting preference is associated with area type.

Step 1. H_0 : commuting preference is independent of area type; H_1 : commuting preference is associated with area type. **(B1)**

Step 2. Expected frequencies: $E(\text{Urban, Car}) = 120 \times 90 / 200 = 54$; $E(\text{Urban, PT}) = 120 \times 70 / 200 = 42$; $E(\text{Urban, Cycle}) = 120 \times 40 / 200 = 24$; $E(\text{Rural, Car}) = 80 \times 90 / 200 = 36$; $E(\text{Rural, PT}) = 80 \times 70 / 200 = 28$; $E(\text{Rural, Cycle}) = 80 \times 40 / 200 = 16$. All $E \geq 5$, so no pooling needed. **(M1A1)**

Step 3. Calculate $X^2 = (40-54)^2/54 + (50-42)^2/42 + (30-24)^2/24 + (50-36)^2/36 + (20-28)^2/28 + (10-16)^2/16 = 196/54 + 64/42 + 36/24 + 196/36 + 64/28 + 36/16 \approx 3.630 + 1.524 + 1.500 + 5.444 + 2.286 + 2.250 = 16.634$. **(M1A1)**

Step 4. $v = (2-1)(3-1) = 2$. Critical value $\chi^2_{2,0.01} = 9.210$. **(B1)**

Step 5. Since $16.634 > 9.210$, X^2 exceeds the critical value. **(M1)**

Answer: There is sufficient evidence, at the 1% level, to conclude that commuting preference is associated with area type. **(A1)**

EXAM TIP

For contingency tables, it is easy to forget that the formula $E = \text{row total} \times \text{column total} / \text{grand total}$ only works for TWO-way tables where independence is genuinely being tested — it is worth deriving this once from basic probability ($P(\text{row}) \times P(\text{column}) \times \text{grand total}$, using independence under H_0) so you can see it isn't an arbitrary rule. It's also worth identifying by eye, after a significant result, which cell(s) contributed most to X^2 (here, Rural/Car and Rural/Cycling), since “comment on the association” follow-up questions are common.

Chapter summary

Idea	Key point
Chi-squared statistic	$X^2 = \sum (O - E)^2 / E$
Pooling rule	Combine categories so every expected frequency is at least 5, then recount categories
Goodness-of-fit d.f.	(categories) – 1 – (parameters estimated)
Contingency table d.f.	(rows – 1)(columns – 1)
Contingency table expected frequency	$E = \text{row total} \times \text{column total} / \text{grand total}$
Decision rule	Reject H_0 if X^2 exceeds the critical value from χ^2 tables

Common exam mistakes

- Calculating v before pooling categories, rather than after (the number of categories used for v must reflect any pooling).
- Forgetting to subtract 1 extra degree of freedom for each parameter estimated from the data.
- Not checking or not pooling when an expected frequency is below 5, or pooling two categories that aren't adjacent (categories must stay in their natural order, e.g. you can pool “3” with “ ≥ 4 ” but not “0” with “ ≥ 4 ”).
- Reading the wrong row/column in χ^2 tables (mixing up significance level and degrees of freedom).
- Concluding “the variables are independent” (accepting H_0) rather than “insufficient evidence of association” when the test is not significant.

You should now be able to

- Carry out a chi-squared goodness-of-fit test, including when a parameter must be estimated from the data.

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- Correctly find degrees of freedom, accounting for both pooling and estimated parameters.
 - Pool categories appropriately so that all expected frequencies are at least 5.
 - Carry out a chi-squared test for association using a contingency table, including finding expected frequencies and degrees of freedom.
 - State conclusions about goodness-of-fit or association clearly, in context.